Thoracoabdominal Movement Patterns in Older Adults

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Introduction

Thoracoabdominal movement reflects the mechanical interaction between the superior thorax (ST), the inferior thorax (IT), and the abdomen (AB) during breathing [1]. Aging can alter thoracic cage mobility, respiratory muscle function, and ventilatory mechanics [2]; however, it remains unclear how the respiratory pattern in older adults performs when multiple kinematic variables are analyzed in an integrated manner[3]. This preliminary study aimed to identify thoracoabdominal movement patterns in older adults and to investigate whether demographic (i.e., age and gender) and anthropometric characteristics (i.e., BMI) are associated with these patterns.

Methods

This exploratory, cross-sectional, observational, preliminary study included 56 older adults, i.e., age above 60 (67.86 ± 5.21 years). Three-dimensional kinematic data of the trunk were obtained using a Vicon system with 12 cameras operating at 100 Hz and a model consisting of 89 reflective markers [1]. Participants performed tidal volume and slow vital capacity maneuvers (SVC) in the standing position. The present preliminary analysis focused on SVC, i.e., the maximum volume of air a person can slowly and completely exhale from the lungs after a deep inhalation.

Thoracoabdominal volume was estimated from the three-dimensional coordinates of the markers using a prism-based model. The percentage of inspiratory contribution from ST, IT, and AB to the total volume was calculated, as well as the coordination between compartments (STxIT, STxAB, and ITxAB) using the phase angle [3].

Percentage data were transformed using logarithmic ratios for compositional variables; therefore, converted into 2 variables (log[ST/IT) and log[IT/AB]) instead of 3. All 5 metrics were standardized using a robust z-score. Uni and multivariate outliers were removed. Cluster analysis was performed using the k-means method. Continuous variables were compared using the Mann-Whitney U test, and categorical variables were compared using Fisher's exact test. Significance level was set at $p < 0.05$.

Results

Two clusters were identified during the vital capacity maneuver (Figure 1). Cluster 1 showed a higher contribution from ST compared to Cluster 2 (34.9 ± 8.94% vs. 27.7 ± 6.22%; $p = 0.011$), while Cluster 2 showed a higher contribution from the IT (30.4 ± 7.19% vs. 19.6 ± 3.48%; $p < 0.001$). Coordination showed significant differences between the clusters for STxIT ($p < 0.001$) and ITxAB ($p < 0.05$).

Age was similar across the clusters (67.9 ± 4.3 vs. 68.05 ± 6.3 years; $p = 0.716$), as well as gender (74.2% vs. 52.0% of women). In contrast, BMI was significantly higher in Cluster 2 (27.6 ± 4.30 vs. 24.8 ± 2.58 kg/m²; $p = 0.030$).

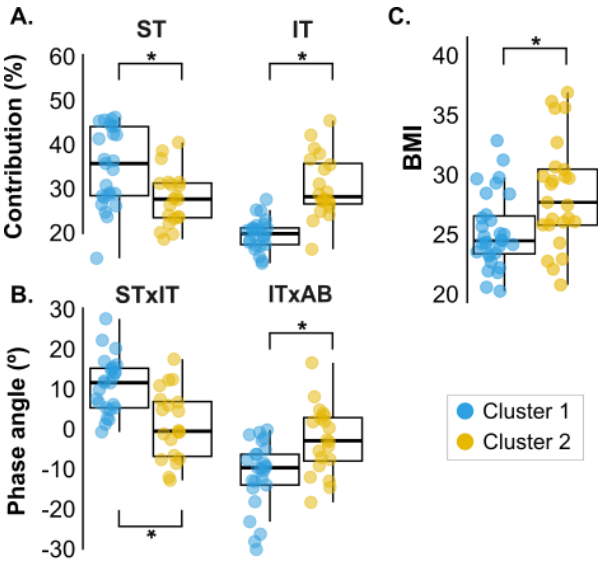


Figure 1: A) Percentage contribution. B) Coordination between compartments. C) BMI. Only significant differences are shown.

Conclusions

This preliminary study identified two distinct thoracoabdominal breathing patterns during the SVC maneuver in older adults. One pattern was characterized by predominant ST contribution and greater intercompartmental asynchrony (Cluster 1), whereas the other showed predominant IT contribution and more coordinated chest wall movement (Cluster 2). BMI was the primary factor differentiating these patterns, highlighting the potential influence of anthropometric characteristics on respiratory mechanics in aging. Future studies should account for these factors when evaluating thoracoabdominal function in older populations.

References

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